DETAILED PROBLEM STATEMENT

Hospital Patient Data Management System Using Doubly Linked List and Queue

Hospitals need an organized way to maintain patient information and manage the order in which patients wait for consultation. Manual record keeping can make it difficult to search for a patient's details, update or remove records, and keep track of the consultation queue. Therefore, a Hospital Patient Data Management System is to be developed in C++ using a Doubly Linked List and a Queue. The system must store each patient's unique ID, name, age, contact details, and reason for visiting. The Doubly Linked List should support inserting new patient records, displaying all stored records, searching by patient ID, updating details, and deleting a record when required. Each linked-list node should contain pointers to both the previous and next nodes, enabling traversal in either direction. A separate Queue should manage patients waiting for consultation. When a patient joins the queue, their ID or reference should be added at the rear; when the doctor is ready, the patient at the front should be called first and removed from the waiting queue, following the First In, First Out (FIFO) rule. The program should provide a clear menu for performing these operations and display suitable messages when a patient is not found or the queue is empty. The system should keep patient-record operations separate from queue operations so that removing someone from the waiting queue does not automatically delete their stored patient record. The project should demonstrate how a Doubly Linked List supports flexible record management and how a Queue supports fair, orderly patient flow. It is intended as an educational prototype and should use sample data rather than real confidential patient information.